ACGT-Net: Adaptive Cuckoo Refinement-Based Graph Transfer Network for Hyperspectral Image Classification

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***Abstract*— Deep learning (DL) has brought many new trends for hyperspectral image classification (HIC). Graph neural networks (GNNs) are models that fuse DL and structured data. Although GNN-based methods have focused on modeling relations, most of them are susceptible to noise, being adverse to capturing hidden correlations from data. Moreover, the existing related approaches typically adopt changeless graph structures, which might lead to poor generalization. To solve the problems mentioned above, this article develops an adaptive cuckoo refinement-based graph transfer network (ACGT-Net) that introduces a meta-heuristic optimization strategy to refine the graph structure. Specifically, we first pretrain a graph convolutional network (GCN) to learn transferable weight parameters. In the undirected graph, nodes are associated with pixels, and edges correspond to similarities between nodes. Afterward, we integrate a cuckoo search strategy (CSS) into the trained GCN to adaptively refine the graph structure. The graph structure refinement (GSR) with the CSS can pay more attention to significant channels by global optimization to improve the generalization of the GNN. Several experiments with real datasets verify the effectiveness and competitiveness of our ACGT-Net compared with other state-of-the-art (SOTA) methods.**

***Index Terms*— Cuckoo search strategy (CSS), graph structure refinement (GSR), hyperspectral image classification (HIC), meta-heuristic optimization, transfer learning (TL).**

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I. Introduction

HYPERSPECTRAL remote sensors are able to acquire spatial details and spectral information from land surfaces, producing a 3-D-cube data for land cover/use research [1], [2], [3], [4]. The 3-D cube is called the hyperspectral image (HSI) cube that has been widely used in many fields over the past decades, including land management, environmental monitoring, and precision agriculture [5], [6], [7]. Since the sensors contain hundreds of narrow channels ranging from visible to near infrared, an imaged pixel can attach a spectral signature with characterized spectral information [8], [9], [10]. Spectral signatures of pixels can be used for discriminating materials in an HSI [11]. Hyperspectral image classification (HIC) can categorize individual pixels in an HSI into a class with a particular label [12], [13], [14]. Over the past several decades, researchers have adopted statistics and machine learning to develop many classical handcrafted methods, such as the spectral analysis [15], logistic regression [16], and wavelet transform [17]. Although HSIs provide a new path for researching land cover/use, high dimensionality with information redundancy causes an unprecedented challenge in classification [18]. The performance of HIC heavily depends on comprehensive low-dimensional feature representation [19], [20], [21].

Feature representation is a typical way to relieve the Hughes phenomenon and explore hidden information to enhance HIC performances, such as the spatial translation invariant wavelet [22], local binary patterns [23], and weighted feature fusion [24]. These methods significantly improve classification performance and demonstrate the role of feature representation [25]. Although conventional approaches are able to acquire satisfactory accuracies in simple scenarios, they usually rely on experiences to extract shallow features [26], [27], [28].

In recent years, the development of deep learning (DL) has inspired much new research in the remote sensing (RS) community, and many deep neural networks are used for achieving HIC, including autoencoder networks [29], residual networks [30], transformer networks [31], [32], [33], and convolutional neural networks (CNNs) [34]. Although the existing DL-based methods can achieve SSC, enhancing generalizability still is an actual demand for facing com-

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plex scenarios [35]. Recently, graph neural network (GNN) applications have become one of the most popular topics in the RS community and have enlightened many new HIC approaches, such as the spectral–spatial graph convolutional network (S^2GCN) [36], multiscale dynamic GCN [37], nonlocal GCN [38], graph attention network [24], minibatch GCN (miniGCN) [39], and spatial–spectral adaptive probability GCN (SSAPGCN) [40]. These aforementioned GNN-based methods focus on modeling relations as opposed to most DL-based approaches. However, the performance of these methods is susceptible to noise, being adverse to exploring hidden correlations from data. In addition, related works usually adopt changeless graph structures, which might not automatically rectify misidentifications associated with properties during the learning stage [41].

Swarm intelligence algorithms (SIAs) use meta-heuristic strategies to solve complex optimization problems, and they typically follow guided random searches to enhance robustness and generalization [42], [43]. Over the past few years, meta-heuristic optimizations have been applied for processing HSIs, such as the particle swarm optimization [44], [45] and ant colony optimization [46], [47]. According to [48], meta-heuristic mechanisms can be utilized for optimizing graph structures via swarm intelligence procedures, improving the availability in feature representations. The cuckoo search strategy (CSS) simulates cuckoo's obligate brood-parasitic behavior, which is also an SIA for implementing optimization tasks [49]. Compared with other SIAs, the CSS is more suitable for optimizing complex models, because its mechanism has fewer parameters to fine-tune [50], [51], [52].

Transfer learning (TL) is an effective strategy that can apply the gained knowledge to conduct a different but related problem [53]. In addition, for classification tasks, TL can mitigate the dependence on large amounts of training samples [54]. Graph TL is a special mechanism that is developed for meeting node classification with structured data [55]. The mechanism usually contains a supervised/semisupervised learning and a domain adaptation [56]. The former uses labeled samples to achieve node embeddings, and the latter facilitates knowledge transfer. Many studies also show that graph TL has great development potential for node classification [55].

In GNN applications, information redundancy and noise may cause structural perturbation, limiting the classification performances of GNNs [57]. Since HSIs cannot avoid information redundancy, they cannot be perfect graph data. Therefore, structural perturbation is an unavoidable problem when using graph model-based networks on HSIs. To address the aforementioned problem, we integrate the CSS into graph TL to develop an adaptive cuckoo refinement-based graph transfer network (ACGT-Net) for HIC. In our ACGT-Net, assuming that each node is associated with a pixel, we adopt similarities between nodes to denote edges and utilize local correlations to construct a graph structure for data preliminarily. Considering that the graph may be corrupted by information redundancy, we employ the CSS to adaptively adjust nodes and edges to refine the graph structure by a meta-heuristic optimization. The CSS-based graph structure refinement (CSS-based GSR) can help the deep network pay more attention to the significant features, which decreases redundant information under preserving intrinsic correlations. Moreover, we develop a hybrid domain adaptation to implement graph TL. Note that, in our ACGT-Net, features only refer to features of nodes. The GCN in the upstream task is as source network and can output primary features. After preclassification by the trained GCN, we design a metrological window to constitute the target domain. In the downstream task, the refined GCN is as target network and can output intermediate features. Finally, we fuse primary and intermediate features as multimodal fusional features and utilize a softmax layer to classify nodes and label pixels. Unlike other GNN-based HIC methods, our ACGT-Net accommodates a meta-heuristic optimization model in the deep graph model, achieving a multimodal joint representation that can enhance the generalization. The main contributions of this work are summarized in the following three points.

1) The idea of meta-heuristic optimization is integrated into GCN to adaptively refine graph structure to enhance classification performance.
2) The CSS-based GSR can mitigate structural perturbations to help GNN better accommodate HSI.
3) The developed graph TL can reduce the dependency of model performance on training samples, which can further enhance the generalization of our approach.

The remainder of this article is organized as follows. Section II briefly reviews two related works: CSS and TL. In Section III, we describe our ACGT-Net for HIC in detail. Afterward, Section IV evaluates our ACGT-Net by drawing comparisons with other state-of-the-art (SOTA) classification approaches. Finally, Section V concludes this article with some remarks and provides plausible hints at our future research lines.

II. Related Works

This section mainly provides theoretical perspectives of CSS and TL.

A. Cuckoo Search Strategy

The CSS is a meta-heuristic algorithm that simulates cuckoo breeding to achieve optimization tasks [49]. In nature, cuckoos are brood-parasitic birds that do not build their own nests and lay their eggs in the nests of other host birds. Cuckoos may remove others' eggs to increase the hatching probability of their eggs [50]. Female cuckoos may imitate in color and shape of the eggs of the selected host birds, aiming to prevent hosts from discovering brood-parasitic behavior. For cuckoo species, the choice of nest directly affects the survival rate of the next generation. According to [50], the simulation of the female cuckoo behavior follows three idealized rules.

1) Each cuckoo lays one egg and dumps its egg in a random host nest at a time.
2) In randomly selected nests, the nests with the best hatchability will be reserved for the next generations.
3) The number of available host nests is fixed in the assumption, and the cuckoo's egg may be discovered by the host with a probability ρ, with $\rho \in [0, 1]$.

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If a host bird senses the eggs are not its own, it will throw them away or abandon the nest. For the CSS, the third rule needs to be approximated by considering ρ of all available nests [51]. The fitness of a solution is proportional to the objective function. The CSS assumes that each egg in a nest represents a solution [49]. In each round of next generations, the best generations will continue to reproduce in the current nest (the current optimal solution), and the rest of the generations need to find new nests. In the CSS, the updates are based on the Lévy flight [50], [58].

B. Transfer Learning

Traditional machine learning is able to predict results or implement tasks by statistical regression models, where the labeled and unlabeled data are satisfied with the uniform distribution assumption [59], [60]. Conversely, TL allows different tasks and distributions in the whole learning procedure, which can solve some new problems faster by utilizing previously obtained knowledge [54]. TL usually involves two particular concepts that are domain and task, respectively. Domains typically contact data and feature spaces, while tasks usually need loss functions to guide [55]. TL can acquire knowledge from a source domain (or multiple source domains) and applies the knowledge to a target domain, aiming to transfer available information to the target task as more as possible [61]. Furthermore, TL rarely uses the symmetric structure to connect the source and target tasks [56].

Considering the source and target domains, TL has developed three ramifications: 1) inductive; 2) transductive; and 3) unsupervised TLs [53]. For the inductive TL, regardless of whether the source and target domains are the same, there are differences between the target task and the source task. The transductive TL assumes that the source and target tasks are consistent, while the source domain differs the target domain [62]. The unsupervised TL is similar to the inductive TL, while it usually adopts unsupervised ways to complete the task in the target domain [63]. In addition, considering specific procedures, TL contains different types, such as parameter-based TL, relation-based TL, feature-based TL, and instance-based TL. The parameter-based TL can be achieved by finding parameters shared between source and target domains [64]. The relation-based TL can acquire the relationships between the source and target domains to transfer the knowledge [65]. The feature-based TL can map the samples of source and target domains into feature space, which can reduce the discrepancy between the marginal and conditional distributions of the source and target domains [66]. The instance-based TL usually involves certain rules that can reuse partial data for learning knowledge [53].

As a branch of TL, graph TL typically combines domain adaptation and GNN to achieve node embedding [55]. Domain adaptation represents an ability that can transfer one or more source domains to a different (but related) target domain. For the classification task, domain adaptation can relieve negative disturbances when transferring knowledge from a source domain to a target domain [53]. Moreover, the domain adaptation on graph-structured data usually meets the assumption that the samples are independent and identically distributed in all domains [55].

III. Proposed ACGT-Net

The proposed ACGT-Net depends on three main components: construction of a graph, hybrid domain adaptation, and CSS-based GSR. The framework of ACGT-Net refers to an upstream task and a downstream task, as illustrated in Fig. 1. In the upstream task, we employ a spatial weighted mean filter (SWMF) [67], [68] to capture local characteristics, aiming to assign initial correlations for nodes. The data processed by the SWMF have an initial graph structure, because adjacency weights between nodes are reassigned by aggregating the spatial information under a 2-D Euclidean distance. For the graph transfer network, an HSI processed by the SWMF is regarded as the source domain data and is used for pretraining a GCN. In the upstream task, the latent features from the last GCN layer are defined as the primary features. The proposed ACGT-Net adopts a hybrid domain adaptation to connect the upstream and downstream tasks. Moreover, we designed a metrological window that can utilize preclassification and classified ratios to acquire target domain data. In the downstream task, the CSS can adjust nodes and edges to refine the graph structure for GCN. The GCN in the upstream task is regarded as the source network, and the GCN in the downstream is the target network. The weight parameters of the source network are transferred to the target network before implementing CSS-based GSR. In the downstream, the latent features from the last GCN layer are denoted as the intermediate features. Note that the CSS-based GSR is unsupervised, while the target network in the downstream task still adopts supervised training. Finally, primary and intermediate features are fused as multimodal features into a softmax layer for classifying nodes.

A. Initialization of Graph

Before achieving TL, the original hyperspectral data convert to accommodate the graph structure model. Specifically, we employ the SWMF to obtain local spatial characteristics for assigning initial adjacency weights between nodes. Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_p]^\top = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_d] \in \mathbb{R}^{p\times d}$ be an HSI, where $\mathbf{x}_i \in \mathbb{R}^{d\times 1}$ denotes a pixel and $\boldsymbol{X}_j$ represents a band. Before implementing the SWMF, we normalize nodes to [0, 1]

$$x_j = \frac{x_j - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

where x_j is the jth element in $\boldsymbol{X}_j$, $j = 1, 2, \ldots, d$. Herein, $x_{\max}$ and $x_{\min}$ are the minimum and maximum elements in $\boldsymbol{X}_j$, $j = 1, 2, \ldots, d$, respectively. The purpose of (1) is that constrains spectral reflectances to a reasonable range. After implementing the SWMF, the observed HSI can be transformed into structured data that can aggregate intrinsic connections. Let $\tilde{\mathbf{x}}_i$ be a structured data obtained from the SWMF, the implementation of the SWMF can be represented as follows:

$$\tilde{\mathbf{X}} \Leftarrow \tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i + \sum_{s=1}^{N^2-1} a_{(i,s)}\mathbf{x}_s}{1 + \sum_{s=1}^{N^2-1} a_{(i,s)}} \tag{2}$$

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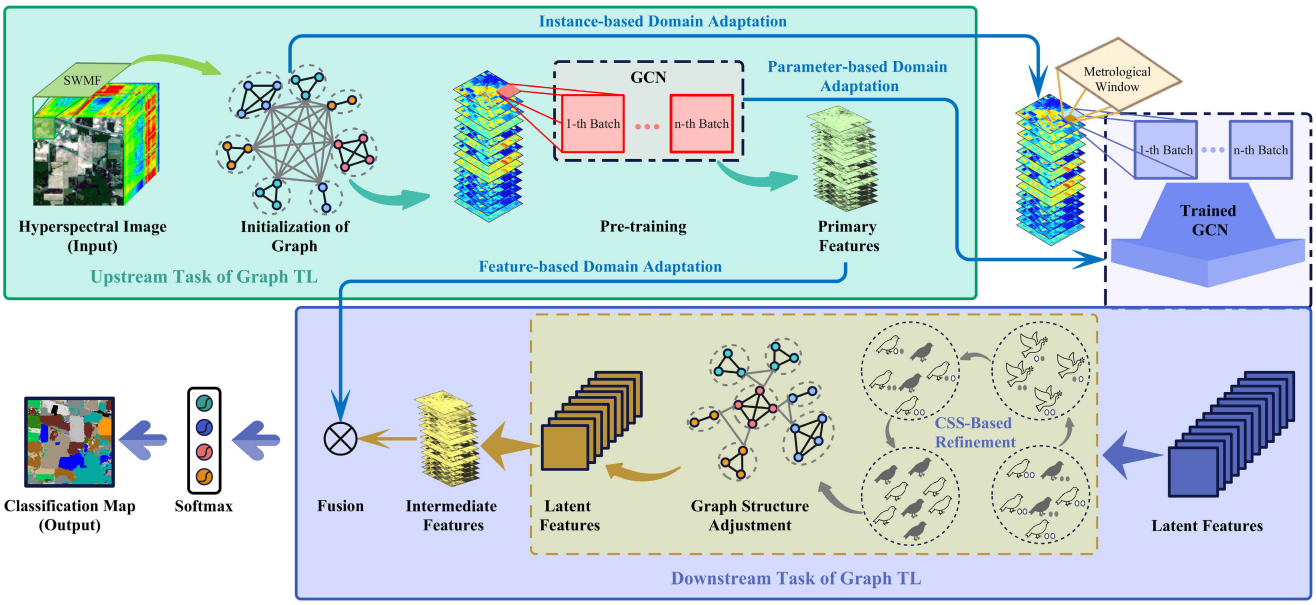


Fig. 1. Flowchart of the proposed ACGT-Net. The framework of ACGT-Net involves an upstream task and a downstream task, and the newly proposed approach mainly contains three components: construction of a graph, hybrid domain adaptation, and CSS-based GSR. The upstream task can construct a graph and pretrains a GCN, while the downstream achieves the hybrid domain adaptation and CSS-based GSR. In this work, the hybrid domain adaptation contains instance-based, feature-based, and parameter-based procedures. Finally, primary and intermediate features are fused as multimodal features into a softmax layer for classifying nodes and labeling pixels.

where N is a territorial hyperparameter of the SWMF and is an odd number and $\mathbf{x}_s$ represents a neighboring node of $\mathbf{x}_i$, $s = 1, 2, \ldots, N^2 - 1$. In (2), $a_{(i,s)}$ represents a similarity weight obtained by a radial basis function (RBF)

$$a_{(i,s)} = \exp\left(-\alpha \|\mathbf{x}_i - \mathbf{x}_s\|_2^2\right) \tag{3}$$

where α determines the width of a Gaussian kernel. The aforementioned procedure can obtain local correlations for nodes. The undirected graph can be written as $\mathcal{G} = (\tilde{\mathbf{X}}, \mathbf{E})$, where $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \ldots, \tilde{\mathbf{x}}_p]^\top \in \mathbb{R}^{p\times d}$ represents all nodes and $\mathbf{E} \in \mathbb{R}^{p\times p}$ is the adjacency matrix that can define all edges. Considering computational burdens generated from DL, we adopt a batchwise scheme to train the ACGT-Net. In this work, we set c is the number of nodes in a batch, and assume n is the number of batches. Each subgraph associated with a batch can be defined as $\mathcal{G}_k = (\tilde{\mathbf{X}}_k, \mathbf{E}_k)$, $k = 1, 2, \ldots, n$, where $\tilde{\mathbf{X}}_k \in \mathbb{R}^{c\times d}$ is node matrix of the subgraph and $\mathbf{E}_k \in \mathbb{R}^{c\times c}$ represents the edges of the subgraph. Each element $e_{(k,i,t)}$ in $\mathbf{E}_k$ can be acquired by an RBF, $t = 1, 2, \ldots, c$. The acquisition with respect to $\mathbf{E}_k$ can be formulated as follows:

$$\mathbf{E}_k \Leftarrow e_{(k,i,t)} = \exp\left(-\beta \left\|\tilde{\mathbf{x}}_{(k,i)} - \tilde{\mathbf{x}}_{(k,t)}\right\|_2^2\right) \tag{4}$$

where β is also utilized to determine the width of a Gaussian kernel, and $\tilde{\mathbf{x}}_{(k,i)}$ and $\tilde{\mathbf{x}}_{(k,t)}$ represent nodes in the kth subgraph, $i \neq t$.

B. Hybrid Domain Adaptation

In the proposed ACGT-Net, we adopt a hybrid domain adaptation to achieve knowledge transferring. The hybrid domain adaption involves three procedures: instance-based, feature-based, and parameter-based procedures. The instance-based procedure can transfer knowledge information from source domain data to target domain data; the feature-based procedure retains the ability of the source GCN by transferring primary features from a source GCN to a target GCN; the parameter-based procedure means that the source GCN provides initial weights parameters to a target GCN.

In the kth batch, $\tilde{\mathbf{X}}_k$ represents source domain data, and $\tilde{\mathbf{X}}'_k$ denotes target domain data. Considering nodes and edges, the graph in a source GCN can be represented as $\mathcal{G}_k = (\tilde{\mathbf{X}}_k, \mathbf{E}_k)$, and the graph in a target GCN can be denoted as $\mathcal{G}'_k = (\tilde{\mathbf{X}}'_k, \mathbf{E}'_k)$. Assuming that $\mathbf{B}_k \in \mathbb{R}^{c\times c}$ is the degree matrix of $\mathbf{E}_k$, $\mathbf{B}_k$ can acquired as the following equation:

$$\mathbf{B}_k \Leftarrow b_{(k,i,t)} = \sum_{t=1}^{c} e_{(k,i,t)} \tag{5}$$

where $\mathbf{B}_k$ is a diagonal matrix and $b_{(k,i,t)}$ is an element of $\mathbf{B}_k$. According to [69], if the GCN has g GCN layers, the latent features in a GCN layer can be represented as follows:

$$\tilde{\mathbf{H}}_k^{(l+1)} = \mathrm{ReLU}\left(\tilde{\mathbf{B}}_k^{(l)^{-\frac{1}{2}}} \tilde{\mathbf{E}}_k^{(l)} \tilde{\mathbf{B}}_k^{(l)^{-\frac{1}{2}}} \tilde{\mathbf{H}}_k^{(l)} \mathbf{W}_k^{(l)}\right) \tag{6}$$

where $\mathrm{ReLU}(\cdot)$ is the activation function and $\mathbf{W}_k^{(l)}$ is the weight parameters of the lth layer, $l = 0, 1, \ldots, g-2$. Note that $\tilde{\mathbf{H}}_k^{(0)} = \tilde{\mathbf{X}}_k$ when $l = 0$. In the GCN, each parameter matrix $\mathbf{W}_k^{(l)}$ is learnable. Herein, $\tilde{\mathbf{B}}_k^{(l)}$ and $\tilde{\mathbf{E}}_k^{(l)}$ are given by

$$\begin{cases} \tilde{\mathbf{E}}_k^{(l)} = \mathbf{E}_k^{(l)} + \mathbf{I} \\ \tilde{\mathbf{B}}_k^{(l)} \Leftarrow \tilde{b}_{(i,t)} = \sum_{t=1}^{c} \tilde{e}_{(i,t)} \end{cases} \tag{7}$$

where $\tilde{b}_{(k,i,t)}$ and $\tilde{e}_{(k,i,t)}$ are two elements in $\tilde{\mathbf{B}}_k^{(l)}$ and $\tilde{\mathbf{E}}_k^{(l)}$, respectively, and $\mathbf{I}$ is an identity matrix. We still adopt the

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RBF in (4) to update edges

$$\mathbf{E}_k^{(l)} \Leftarrow e_{(k,i,t)} = \exp\left(-\beta\left\|\tilde{\mathbf{h}}_{(k,i)}^{(l)} - \tilde{\mathbf{h}}_{(k,t)}^{(l)}\right\|_2^2\right) \tag{8}$$

where $\tilde{\mathbf{h}}_{(k,i)}^{(l)}$ and $\tilde{\mathbf{h}}_{(k,t)}^{(l)}$ are two row vectors in $\widetilde{\mathbf{H}}_k^{(l)}$ and they represent the latent features of the ith node and tth, respectively. To achieve classification, we employ the softmax activation to acquire class probabilities of nodes. The output features of the gth GCN layer can be formulized as follows:

$$\widetilde{\mathbf{H}}_k^{(g)} = \text{Softmax}\left(\widehat{\mathbf{E}}_k^{(g-1)}\widetilde{\mathbf{H}}_k^{(g-1)}\mathbf{W}_k^{(g-1)}\right) \tag{9}$$

where $\widehat{\mathbf{E}}_k^{(g-1)}$ is the renormalized adjacency matrix and $\widehat{\mathbf{E}}_k^{(g-1)} = \tilde{\mathbf{B}}_k^{(g-1)^{-(1/2)}}\tilde{\mathbf{E}}_k^{(g-1)}\tilde{\mathbf{B}}_k^{(g-1)^{-(1/2)}}$. After implementing all batches, all outputs are used for constituting the primary features $\mathbf{H}^{(g)} \in \mathbb{R}^{p\times\nu}$, where ν represents the number of classes

$$\mathbf{H}^{(g)} = \left[\widetilde{\mathbf{H}}_1^{(g)}, \widetilde{\mathbf{H}}_2^{(g)}, \ldots, \widetilde{\mathbf{H}}_n^{(g)}\right]. \tag{10}$$

The primary features will be used for obtaining fusional features in the downstream task. In the upstream task, the number of training samples is μ. Assuming that $\{\mathbf{h}^{(g)}, \mathbf{y}\}$ denotes a training sample, $\mathbf{h}^{(g)}$ is a sample vector, and $\mathbf{y} \in \mathbb{R}^{\nu\times 1}$ is the one-hot label vector

$$\mathcal{L}_1 = -\sum_{m\in\Lambda_\mu}\sum_{\gamma=1}^{\nu} y_{(m,\gamma)}\log\left(h_{(m,\gamma)}\right) \tag{11}$$

where $y_{(m,\gamma)}$ is an indicator variable that labels node to 1 or 0, Λ_μ is a set of node indices, and $h_{(m,\gamma)}$ is the probability of the mth node belonging to the γth class. Since a node associates a pixel, we can accomplish the preclassification on an image by node classification.

After completing the preclassification, we design a metrological window to achieve the instance-based procedure. Concretely, the metrological window can count the number of all labels and calculate the proportion of each land-cover type to the total labels. The proportion is a classified ratio that represents the percentage of each class in the window. In metrological windows, we set a threshold-based measurement for optimizing training samples. If a classified ratio with respect to a land-cover type is less than the threshold, the related nodes will be regarded as indistinguishable samples. Afterward, we remove these indistinguishable samples from the aforementioned training samples, aiming to make target domain data different from source domain data and mitigate the negative effect of source domain data. Although the training samples in the target GCN are still chosen from $\tilde{\mathbf{X}}$, the training sample sets has changed via the metrological window. Considering the change, the instance-based procedure can be represented as follows:

$$\left\{\tilde{\mathbf{x}}_i\right\}_{q=1}^{\phi} \xleftarrow[\text{Indistinguishable Samples}]{\text{Without}} \left\{\tilde{\mathbf{x}}_i\right\}_{r=1}^{\mu} \tag{12}$$

where the number of samples is ϕ in the downstream task, $\phi < \mu$. In the proposed approach, weight parameter matrix $\mathbf{W}_k$ of each GCN layer is obtained by learning. The parameter-based procedure can provide initial weights to the target GCN from the source GCN. In the target GCN, the $(g+1)$th GCN layer is set as the prediction layer. Furthermore, only the weight parameters of the $(g+1)$th GCN layer are variable with learning. Fig. 2 illustrates the information propagation in the proposed ACGT-Net.

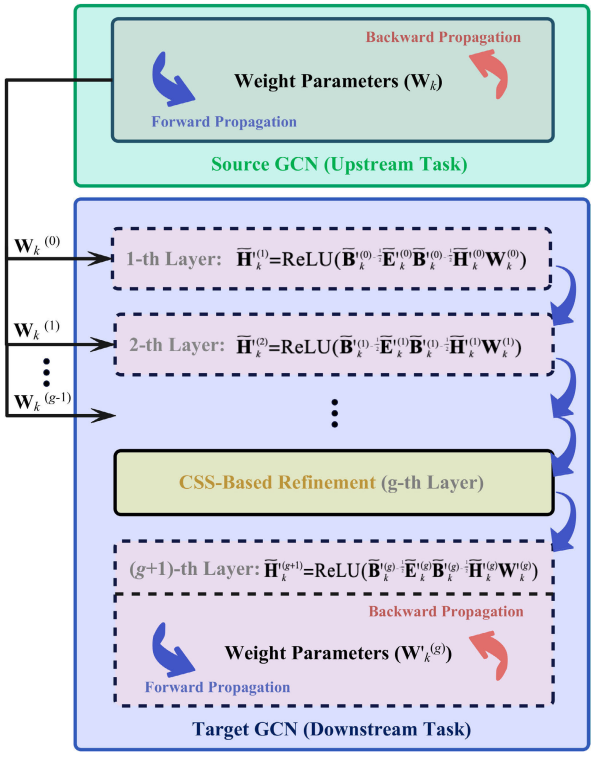


Fig. 2. Illustrations of weight updates and information propagations. The source GCN updates all parameters and conducts forward and backward propagations, while the target GCN only updates weight parameters and implements backward propagation in the $(g+1)$th layer.

C. CSS-Based GSR

The CSS-based GSR also adopts a cross entropy to design its loss function for achieving meta-heuristic optimization. Considering the favorable expansibility, the CSS collaborates with the deep network and can lead the attention of the downstream task to achieve a particular node embedding. Specifically, the RBF in (8) acquires the attention of edges, and the CSS can enable the GCN to pay more attentions to the significant channels by global optimization. The particular node embedding is beneficial to represent the internal relationships between nodes better. Moreover, the CSS-based GSR introduces a global optimization model, which can further enhance the generation of our ACGT-Net. With implementing CSS, the gth layer in the target network can update nodes and edges.

According to [49], [50], and [51], the CSS utilizes Lévy flights to conduct the random walk, aiming to explore more search spaces to exclude local optima as much as possible. For a Lévy flight, the random step length follows a Lévy distribution with a power law that can be expressed as follows:

$$\text{Lévy}(\delta) \sim \frac{\psi}{|\varepsilon|^{\frac{1}{\delta}}} \tag{13}$$

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where Lévy(δ) denotes a random search, $0 < \delta \leq 2$. ψ and ε are two random that are from normal distributions [70], [71]

$$\begin{cases} \psi \sim \text{Normal}\left(0, \sigma_\psi^2\right) \\ \varepsilon \sim \text{Normal}\left(0, \sigma_\varepsilon^2\right) \end{cases} \tag{14}$$

where σ_ψ and σ_ε are standard deviations and can be represented as follows:

$$\begin{cases} \sigma_\psi = 1 \\ \sigma_\varepsilon = \left(\dfrac{\Gamma(1+\delta)\sin\left(\frac{\pi\delta}{2}\right)}{\Gamma\left(\frac{1+\delta}{2}\right)\delta 2^{(\delta-1)/2}} \right)^{\frac{1}{\delta}} \end{cases} \tag{15}$$

where $\Gamma(\cdot)$ represents a gamma function. Although the original CSS was proposed to solve optimization problems under continuous spaces, the related processes can be converted into discrete combinatorial optimization for processing HSIs. In the ACGT-Net, we regard the location of host nests as the optimal solution of the feature selection when implementing CSS. The feature selection will further reduce the dimensionality of previous latent feature. Each feature component will be marked to a binary code (1 or 0). Significant feature components will be marked as 1, while insignificant components will be marked as 0. We adopt a logistic function as the activation function, and the features will be to the range of [0, 1] for measuring significances of components. The logistic function is employed to calculate significance scores of feature components. Considering $\mathbf{H}_k^{(g)} = [\mathbf{h}_1, \mathbf{h}_1, \ldots, \mathbf{h}_r]^\top$, the logistic function can be written as the following formula:

$$\text{Score}\left(\mathbf{h}_{j'}\right) = \frac{1}{1 + \exp\left(-\mathbf{h}_{j'}\right)} \tag{16}$$

where r represents dimension, $j' = 1, 2, \ldots, r$. Corresponding to each feature component, we set a probability ρ within the range of [0, 1]. If ρ is less than the significance score, the feature component will be associated with 1. Otherwise, it will be associated with 0. The algorithm can be denoted as follows:

$$\text{Mark}\left(\mathbf{h}_j\right) = \begin{cases} 1, & \rho < \text{Score}\left(\mathbf{h}_j\right) \\ 0, & \rho > \text{Score}\left(\mathbf{h}_j\right). \end{cases} \tag{17}$$

In the implementation of the CSS, a generation corresponds to a iteration. Each feature component will be assigned a mark (1 or 0) after completing an iteration. All initial marks are set to 0. With the iterative optimization, some marks will be changed to 1. Although the logistic function in (16) is bounded and differentiable, the CSS only can update the binary code but not adjust the weight parameters. Let l represent a generation of the CSS. Following the research in [50], the location of the host nest will be updated as follows:

$$\mathbf{H}_k^{(g,(l+1))} = \mathbf{H}_k^{(g,l)} + \xi \odot \text{Lévy}(\delta) \tag{18}$$

where $\odot$ represents the Hadamard product and ξ is the step size that is related to the scales of interests. We assume that $\mathbf{H}_k^{\text{best}}$ is the best solution at present. The step size can be denoted as follows:

$$\xi = \xi_0 \left(\mathbf{H}_k^{(g,l)} - \mathbf{H}_k^{\text{best}} \right) \tag{19}$$

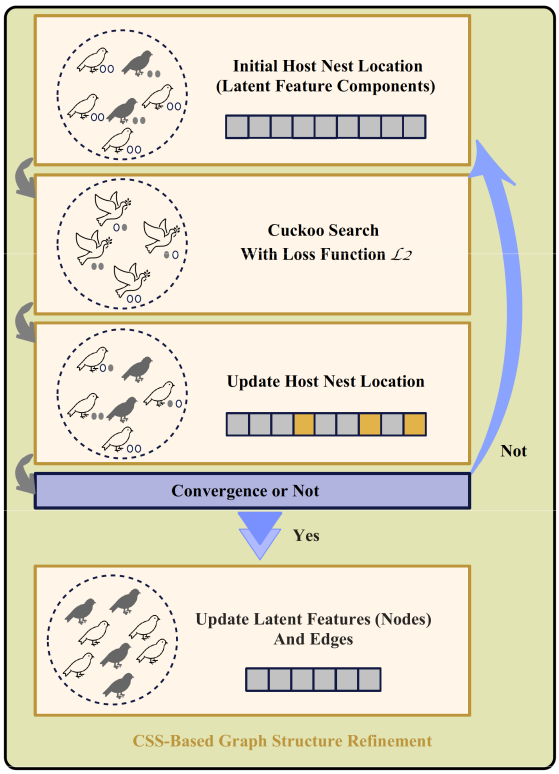


Fig. 3. Illustration of CSS-based GSR. The CSS can enable the target GCN to pay more attentions to the significant features. The gray squares represent 1, and the goldenrod squares represent 0. After completing the CSS, we only reserve the latent feature components corresponded to marks with 1.

where ξ_0 denotes a scaling factor and is empirically set to 0.01, according to [49] and [71]. Introducing (13) and (19) to (18), the location of the host nest will conduct the following update:

$$\mathbf{H}_k^{(g,(l+1))} = \mathbf{H}_k^{(g,l)} + \xi_0 \frac{\psi}{|\varepsilon|^{\frac{1}{\delta}}} \left(\mathbf{H}_k^{(g,l)} - \mathbf{H}_k^{\text{best}} \right). \tag{20}$$

Afterward, the CSS-based GSR still adopt (8) to update the edges in the target network. In the implementation of the CSS, we should not only concern significant feature components but classification performance. Therefore, the loss function of the CSS is set as follows:

$$\mathcal{L}_2 = \mathcal{L}_1\left(1 + \zeta \mathcal{F}\left(\mathbf{H}^{(g)}\right)\right) \tag{21}$$

where $\mathcal{F}(\cdot)$ is a statistical function that can acquire the vertical dimension of matrix and ζ is a balance coefficient with being set to 0.001. Fig. 3 illustrates the related processes of the CSS-based GSR. After completing the CSS, we only reserve the latent feature components corresponded to one mark.

D. HIC With Joint Representation

In the downstream task, the last GCN layer is the $(g+1)$th layer. The new training sample set obtained from (12) can be used to train the $(g+1)$th GCN layer, while the parameters in other GCN layers inherit from the source GCN. Analogously, we still adopt the batchwise scheme to train the target GCN. In the refined GCN, the output of the last GCN layer can be denoted as follows:

$$\widetilde{\mathbf{H}}_k'^{(g+1)} = \text{ReLU}\left(\tilde{\mathbf{B}}_k'^{(g)^{-\frac{1}{2}}} \tilde{\mathbf{E}}_k'^{(g)} \tilde{\mathbf{B}}_k'^{(g)^{-\frac{1}{2}}} \widetilde{\mathbf{H}}_k'^{(g)} \mathbf{W}_k'^{(g)} \right) \tag{22}$$

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where $\widetilde{\mathbf{H}}_k'^{(g+1)}$ represents the node features obtained from the $(g+1)$th GCN layer. Herein, $\tilde{\mathbf{B}}_k'^{(g)}$ and $\tilde{\mathbf{E}}_k'^{(g)}$ are a diagonal matrix and edges, respectively. All batches will provide the outputs to constitute the intermediate features

$$\mathbf{H}'^{(g+1)} = \left[\widetilde{\mathbf{H}}_1'^{(g+1)}, \widetilde{\mathbf{H}}_2'^{(g+1)}, \ldots, \widetilde{\mathbf{H}}_n'^{(g+1)}\right] \tag{23}$$

where $\mathbf{H}'^{(g+1)} \in \mathbb{R}^{p\times v}$ represents the intermediate features. Note that, the acquisition of fusional features has two paths, containing the additive and multiplicative ways. In the target network, we still employ the softmax activation to obtain class probabilities of nodes. Let $\mathbf{F}_{\text{ADD}}$ and $\mathbf{F}_{\text{MUL}}$ be two fusional feature sets obtained by additive and multiplicative ways, respectively

$$\begin{cases} \mathbf{F}_{\text{ADD}} = \text{Softmax}\left(\mathbf{H}^{(g)} \oplus \mathbf{H}'^{(g+1)}\right) \\ \mathbf{F}_{\text{MUL}} = \text{Softmax}\left(\mathbf{H}^{(g)} \odot \mathbf{H}'^{(g+1)}\right) \end{cases} \tag{24}$$

where $\oplus$ represents the elementwise addition and uses the cross entropy to construct the loss function

$$\mathcal{L}_3 = -\sum_{m'\in\Lambda_\phi}\sum_{\gamma=1}^{v} y_{(m',\gamma)} \log\left(f_{(m',\gamma)}\right) \tag{25}$$

where $y_{(m',\gamma)}$ is an indicator variable, Λ_ϕ represents the set of node indices, and $f_{(m',\gamma)}$ represents the output corresponding to the m'th node. Finally, we will assign labels to all pixels via the softmax layer. A pseudocode of our ACGT-Net is given in Algorithm 1.

Algorithm 1 Pseudocode of the Proposed ACGT-Net

Input: Data $\mathbf{X}$; Labels of training sample.
Output: Labels of test samples.

```
Acquire local correlations by (1) and (2);
Obtain initial edges E_k by (4).
for Epochs do
    for Batches do
        Obtain weight parameters W_k by source network; Acquire latent features H̃_k by (6);
        Update edges E_k by (8).
    end
    Constitute primary features H^(g) by (10);
    Achieve the preclassifcation by a softmax layer.
end
Acquire training samples to downstream task by (12);
for Epochs do
    for Batches do
        repeat
            Calculate significance scores of feature components by (16); Assign marks to feature components by (17); Adjust nodes by (20); Update the edges by (8).
        until Reach the convergence in (21);
        Learn weight parameters W_k'^(g) in the gth layer by the target network;
        Update latent features H'^(g+1) in the (g+1)th layer by (22).
    end
    Constitute the intermediate features by (23);
    Obtain fusional features F_ADD (or F_MUL) by the joint representation in (24);
    Predict labels of test samples by a softmax layer.
end
```

IV. Experimental Results and Analysis

To confirm the effectiveness of our ACGT-Net, we used four metrics to assess classification results, including the following: 1) class accuracy (CA); 2) overall accuracy (OA); 3) average accuracy (AA); and 4) kappa coefficient. Moreover, the competitiveness of the ACGT-Net was evaluated by comparing with other SOTA methods. In this article, the SOTA methods are the aforementioned 3-D CNN [72], GCN [69], S^{2}GCN [36], nonlocal GCN [38], miniGCN [39], and SSAPGCN [40], respectively. The GCN, S^{2}GCN, nonlocal GCN, miniGCN, and SSAPGCN belong to GCN-based methods. All experiments have been implemented on the same hardware platform with a Core I7 CPU and a NVIDIA GeForce RTX 3080 ti GPU. In these experiments, we set 200 epochs to implement all methods. Since the acquisition of fusional features has two ways in (24), we use the ACGT-Net$_{\text{ADD}}$ and the ACGT-Net$_{\text{MUL}}$ to represent additive and multiplicative ways, respectively.

A. Data Description

This article uses four real hyperspectral datasets to test our ACGT-Net. The four test datasets contain the Indian Pines,[1] Pavia University,[2] Xiong'an Matiwan,[3] and Houston University.[4] The Indian Pines, Pavia University, and Houston University datasets are satellite-carried sensors, while the Xiong'an Matiwan dataset was acquired from an unmanned aerial vehicle (UAV) platform. The four real data attach their ground-truth maps for evaluating classification results.

1) Indian Pines Dataset: The Indian Pines dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer sensor in 1992. The observed range locates in northwestern Indiana, USA. The observed image contains 16 classes, and the ground truth has 10 249 labeled pixels. The size of the

TABLE I
Means and Standard Deviations After Ten Runs of Source Model and New Methods. Accuracy Comparison by the Indian Pines Dataset

Evaluation Metrics	SWMF+GCN	ACGT-Net$_{\text{ADD}}$	ACGT-Net$_{\text{MUL}}$
Mean OA (%)	82.02 ±2.52	91.35 ±0.39	91.40 ±0.27
Mean AA (%)	79.65 ±2.14	88.09 ±0.46	89.15 ±0.33
Mean Kappa ×100	81.03 ±1.91	89.72 ±0.32	90.03 ±0.29

[1]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[2]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene

[3]The Xiong'an Matiwan dataset was provided by the State Key Laboratory of Remote Sensing Science, Aerospace Information Research Institute, Chinese Academy of Sciences. It can be downloaded from http://www.hrs-cas.com/a/share/shujuchanpin/2019/0501/1049.html

[4]http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

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observed image is 145 × 145 pixels with 200 bands, and its spectral ranging is 0.20–2.40 μm.

2) Pavia University Dataset: The Pavia University dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, Pavia, Italy, in 2001. The coverage region is located in the University of Pavia data. The data acquired by the ROSIS sensor contain 610 × 340 pixels with 103 bands, and the attached ground truth (42 776 labeled pixels) contains nine land-cover types. The geometric resolution is 1.3 m, and the spectral range is from 0.43 to 0.86 μm.

3) Xiong'an Matiwan Dataset: The Xiong'an Matiwan dataset was acquired by the Full-Spectrum Multimode Imaging Spectrometer sensor on a UAV platform, Hebei, China, in 2019. The observed scenario contains 370 × 440 pixels with 250 bands. The geometric resolution is 0.5 m, and the spectral ranging is 0.40–1.00 μm. The observed region comprises eight land-over types, and 1 36 393 pixels are labeled as the ground truth of the dataset.

4) Houston University Dataset: The Houston University dataset was obtained by the ITRES CASI-1500 sensor, TX, USA, in 2012. The Houston University dataset contains 15 classes and attaches 15 029 labeled pixels as the ground truth. The observed image contains 349 × 1905 pixels with 144 bands. The dataset achieves 2.5-m geometric resolution, and its spectral range is from 0.36 to 1.05 μm.

B. Hyperparameter Analysis

The hyperparameters of our ACGT-Net need to be set empirically. In the experiments, we adopt the Indian Pines dataset to test the effects of hyperparameters. α and β are the hyperparameters of Gaussian kernels in (3), (4), and (8). Utilizing the ground truth, we randomly choose 8% labeled samples from each land-cover class as the training samples, and the number of GCN layers is set to 2. The batch size is set to 128, and N is set to 3, and the size of metrological window is set to 11 × 11, and τ is set to 0.01. Fig. 4 is the sensitivity analysis that illustrates the effects when α and β were set to a range. In Fig. 4, the two hyperparameters are set to [0.1, 0.3, 0.5, 1, 3, 5, 7, 9], and we use OA to evaluate classification results. It can observed that the sensitivities of our ACGT-Net$_{\mathrm{ADD}}$ and ACGT-Net$_{\mathrm{MUL}}$ are similar in Fig. 4. In the subsequent experiments, α and β are set to 0.3.

Considering that the proposed ACGT-Net refers to GCN and TL, two ablation experiments are used for evaluating the newly proposed approaches, as shown in Fig. 5. Since the proposed ACGT-Net adopts the SWMF to acquire local spatial correlations, the SWMF was combined with the GCN to ensure the fairness of the experiment. Note that the "2GCNs" in Fig. 5 represent the transfer framework without the CSS-based GSR. The purpose of "SWMF + GCN" is to test the source model, and the "SWMF + 2GCNs" is used for testing TL architecture. The results of ACGT-Net$_{\mathrm{ADD}}$ and ACGT-Net$_{\mathrm{MUL}}$ indicate the effectiveness of GSR, involving the additive and multiplicative fusion ways in (24), respectively. Fig. 5(b) provides a comparison when using different percentages (from 1% to 8%) to choose training samples, where the number of GCN layers is set to 2. The spider chart in Fig. 5(b) reflects the influence of the training samples on classification results.

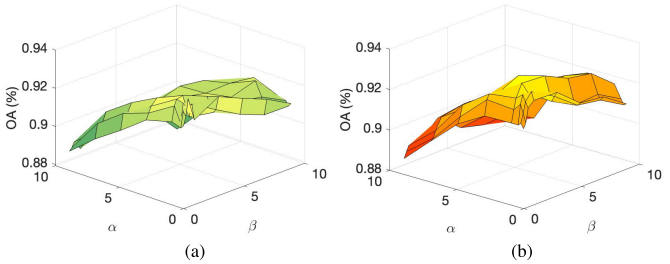


Fig. 4. OAs obtained by the ACGT-Net$_{\mathrm{ADD}}$ and ACGT-Net$_{\mathrm{ADD}}$ under different hyperparameter settings. (a) ACGT-Net$_{\mathrm{ADD}}$. (b) ACGT-Net$_{\mathrm{MUL}}$.

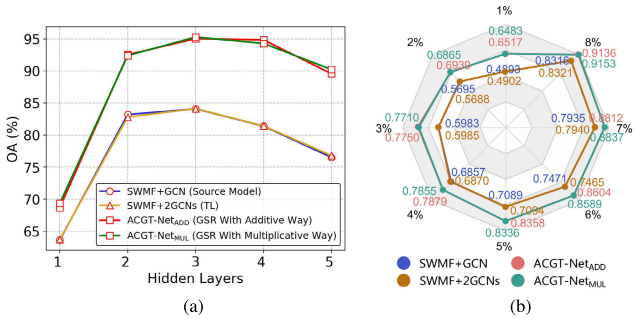


Fig. 5. Ablation experiments for evaluating the proposed ACGT-Net. (a) Changes of OAs along with the number of GCN layers. (b) Kappa coefficients obtained by setting different training samples. The SWMF + GCN and the SWMF + 2GCNs are used for testing the source model and the TL architecture, respectively. The results of ACGT-Net are used for illustrating the effectiveness of GSR. In the comparison of OAs, the number of GCN layer is set as 1–5. In the spider chart of kappa coefficients, we randomly select 1%–8% samples from each ground-truth class as training samples.

In Fig. 5, we can observe that our ACGT-Net model can significantly improve the classification performances. To test the robustness of new methods, we added additive white Gaussian noise (AWGN) into the spectra of the Indian Pines dataset, where the mean and the standard deviation of AWGN are set to 0 and 0.1, respectively. In the noisy condition, we ran the source model and our new methods ten times to test their generalizations. Table I provides means and standard deviations after ten runs of the source model and the new methods. It can be observed that the means and standard deviations of the proposed ACGT-Net are smaller than that of the source model.

For our ACGT-Net, each node corresponds to a pixel, and we use the RBF to compute the similarities between nodes as the edges. If two nodes are very close in feature space, their edge weight is larger and vice versa. For visualization purpose, we use the t-distributed stochastic neighbor embedding (t-SNR) [73] to project nodes into a 2-D feature space. Since the number of nodes in the HSI is too large to show them all, we only use a batch (containing 128 nodes) for the visualization in Fig. 6. The two t-SNE plots show that, after graph refinement, the nodes with the same label are closer, while the nodes with different labels are farther away.

C. Comparative Experiments

1) Experiment With Indian Pines Dataset: In this test, we randomly chosen 8% labeled samples from the ground truth of each land-cover class, and these GCN-based methods

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TABLE II

CLASSIFICATION ACCURACIES OF DIFFERENT APPROACHES, OBTAINED BY USING THE INDIAN PINES DATASET. CAS ARE LISTED FOR COMPARISON. BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	Total	3D-CNN	GCN	S^2GCN	non-local GCN	miniGCN	SSAPGCN	ACGT-Net$_{ADD}$	ACGT-Net$_{MUL}$
CA (%)	Alfalfa	5	49	54	78.15	72.63	75.71	69.93	70.10	**83.73**	74.07	77.78
	Corn-notill	115	1319	1434	84.87	86.82	89.50	85.58	86.54	85.88	**92.05**	91.91
	Corn-mintill	67	767	834	62.71	**85.22**	78.47	84.74	67.63	69.92	85.01	85.13
	Corn	19	215	234	64.53	73.93	90.27	80.01	89.74	96.61	**98.72**	98.29
	Grass-pasture	40	457	497	90.14	92.96	96.93	91.30	93.96	95.83	**97.79**	**97.79**
	Grass-trees	60	687	747	94.78	72.42	98.12	75.76	97.05	99.05	**99.46**	**99.46**
	Grass-pasture-mowed	3	23	26	84.62	69.23	93.20	89.89	**92.31**	78.62	75.00	76.15
	Hay-windrowed	40	449	489	**99.59**	98.57	98.97	98.90	99.38	99.16	**99.59**	**99.59**
	Oats	2	18	20	30.00	45.00	42.76	33.56	35.00	**51.95**	40.00	45.00
	Soybean-notill	78	890	968	75.10	69.94	82.33	80.12	83.06	70.25	**86.47**	86.36
	Soybean-mintill	198	2270	2468	84.24	87.07	75.10	78.83	86.75	84.92	89.55	**89.71**
	Soybean-clean	50	564	614	82.08	51.47	50.38	84.42	77.85	59.53	**89.09**	**89.09**
	Wheat	17	195	212	**100.00**	99.02	**100.00**	99.18	**100.00**	99.03	**100.00**	**100.00**
	Woods	104	1190	1294	95.44	98.38	96.41	**99.07**	98.38	97.29	98.92	**99.07**
	Buildings-Grass-Tress-Drives	31	349	380	66.05	92.63	80.82	93.27	78.42	93.04	94.74	**97.37**
	Stone-steel-towers	8	87	95	90.53	92.64	92.90	93.35	**100.00**	96.47	**100.00**	**100.00**
OA (%)					83.74	84.23	86.44	86.27	87.04	88.16	92.40	**92.54**
AA (%)					80.18	80.50	83.87	83.62	84.76	85.08	88.78	**89.54**
Kappa $\times 100$					81.39	83.13	84.20	84.00	85.24	86.99	91.36	**91.53**

TABLE III

CLASSIFICATION ACCURACIES OF DIFFERENT APPROACHES, OBTAINED BY USING THE PAVIA UNIVERSITY DATASET. CAS ARE LISTED FOR COMPARISON. BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	Total	3D-CNN	GCN	S^2GCN	non-local GCN	miniGCN	SSAPGCN	ACGT-Net$_{ADD}$	ACGT-Net$_{MUL}$
CA (%)	Asphalt	199	6432	6631	92.35	92.34	90.14	92.73	87.20	62.43	92.40	**93.14**
	Meadows	560	18089	18649	**99.06**	94.05	96.87	95.16	93.45	93.89	97.07	96.77
	Gravel	63	2036	2099	70.18	67.79	65.91	65.80	90.52	90.00	91.14	**91.47**
	Trees	92	2972	3064	82.60	93.89	94.00	**95.75**	88.58	60.87	91.61	92.13
	Metal-sheet	41	1304	1345	99.18	99.55	99.49	99.55	99.85	98.72	**100.00**	**100.00**
	Bare-soil	151	4878	5029	81.62	86.62	78.54	90.14	83.52	**92.10**	90.32	90.20
	Bitumen	41	1289	1330	83.08	75.64	76.39	77.61	63.53	86.90	**90.30**	87.44
	Bricks	111	3571	3682	83.46	81.88	65.70	80.94	62.63	83.94	**85.66**	83.65
	Shadows	29	918	947	99.58	96.41	96.59	99.32	**100.00**	52.05	99.68	99.68
OA (%)					91.18	86.15	86.72	90.04	88.75	85.49	**93.83**	93.59
AA (%)					87.90	87.57	84.85	88.56	85.48	79.99	**93.13**	92.72
Kappa $\times 100$					87.42	85.51	82.64	88.05	85.67	77.02	**91.82**	91.51

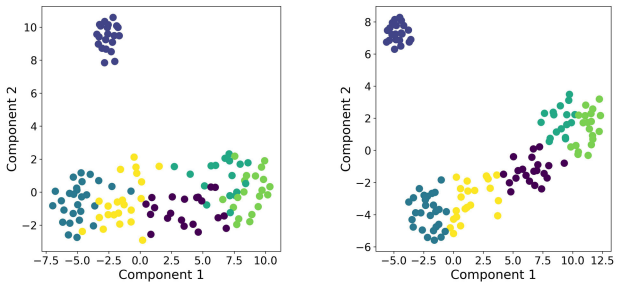


Fig. 6. Plots of obtained feature representations of pixels limited in a batch. (a) Representation of primary features output from a pretrained model's third GCN layer. (b) Representation of intermediate features output from the refined model's third GCN layer.

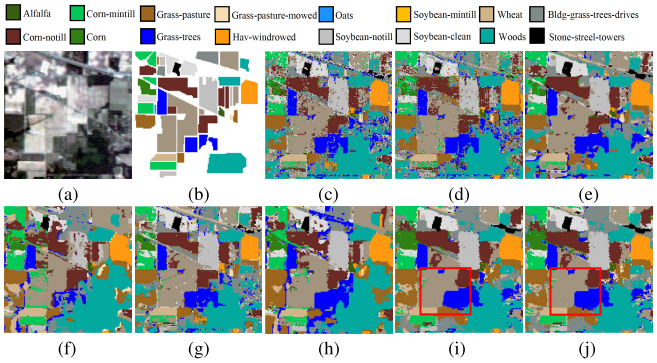


Fig. 7. Classification maps on the Indian Pines dataset, acquired from different HIC approaches. (a) False-color Composition. (b) Ground truth. (c) 3-D CNN. (d) GCN. (e) S^2GCN. (f) Nonlocal GCN. (g) MiniGCN. (h) SSAPGCN. (i) ACGT-Net$_{ADD}$. (j) ACGT-Net$_{MUL}$.

adopt two GCN layers. For the ACGT-Net, we still set the size of batch as 128 and $N = 3$; the size of metrological window is still set to 11, and τ is set to 0.01. Table II is the quantitative comparison and lists the classification results from different HIC methods. In Table II, we can seen that our ACGT-Net significantly improves classification accuracies compared with the single GCN. Moreover, it can be observed in Table II that the ACGT-Net obtained better classification results than other SOTA methods, especially the proposed ACGT-Net$_{MUL}$. Since our ACGT-Net refers to meta-heuristic optimization and TL, it always takes longer computing time. For comparative purposes, Fig. 7 illustrates the classification maps obtained from different approaches. As it can be seen in Fig. 7, the classification maps of our ACGT-Net$_{ADD}$ and ACGT-Net$_{MUL}$ are smoother than that of other SOTA approaches and preserve the boundary details of classes well. Furthermore, in Fig. 7, the results of our ACGT-Net$_{ADD}$ and ACGT-Net$_{MUL}$ are closest to the ground truth compared with other results. The classification results in Table II and Fig. 7 demonstrate the effectiveness and competitiveness of our ACGT-Net.

2) Experiment With Pavia University Dataset: To evaluate the proposed ACGT-Net more comprehensively, we use the Pavia University dataset to implement another comparative experiment. In this experiment, we randomly selected 3% labeled samples from the ground truth of each class as the training samples, and other hyperparameters are set to the same as the Indian Pines experiment. Table III lists the classification

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TABLE IV

CLASSIFICATION ACCURACIES OF DIFFERENT APPROACHES, OBTAINED BY USING THE XIONG'AN MATIWAN DATASET. CAS ARE LISTED FOR COMPARISON. BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	Total	3D-CNN	GCN	S^2GCN	non-local GCN	miniGCN	SSAPGCN	ACGT-Net$_{ADD}$	ACGT-Net$_{MUL}$
CA (%)	Maize	643	20784	21427	94.45	92.54	89.75	91.62	88.30	89.68	92.22	**94.81**
	Pear	1643	53117	54760	93.28	88.28	88.33	89.37	90.18	90.25	93.86	**98.47**
	Soybean	40	1263	1303	90.66	84.16	84.40	91.12	91.29	89.35	93.03	**94.46**
	Populus	290	9376	9666	81.09	85.09	85.52	90.70	97.99	89.74	**98.83**	97.78
	Sophora Japonica	454	14669	15123	96.35	89.17	90.04	95.28	94.69	93.90	94.19	**98.19**
	vegetable Plot	42	1344	1386	96.50	86.53	85.41	90.83	96.88	96.92	96.95	**99.63**
	Grass	279	9013	9292	89.39	84.01	84.73	89.62	89.93	88.61	93.73	**96.46**
	Peach	704	22732	23436	79.48	88.52	89.05	90.30	94.80	95.26	96.49	**98.44**
OA (%)					89.23	89.00	88.63	90.10	91.68	92.05	94.35	**97.65**
AA (%)					90.15	87.29	87.15	91.11	93.01	91.71	94.91	**97.28**
Kappa $\times 100$					84.13	83.44	83.90	88.97	89.22	90.25	92.64	**96.91**

TABLE V

CLASSIFICATION ACCURACIES OF DIFFERENT APPROACHES, OBTAINED BY USING THE HOUSTON UNIVERSITY DATASET. CAS ARE LISTED FOR COMPARISON. BEST RESULTS ARE IN BOLD

	Class Name	Training Samples	Test Samples	Total	3D-CNN	GCN	S^2GCN	non-local GCN	miniGCN	SSAPGCN	ACGT-Net$_{ADD}$	ACGT-Net$_{MUL}$
CA (%)	Healthy-Grass	198	1053	1251	82.73	84.14	77.93	80.39	**98.09**	81.33	84.67	83.96
	Stressed-grass	190	1064	1254	**99.14**	99.08	80.29	85.54	89.72	84.56	83.10	83.10
	Synthetic-Grass	192	505	697	99.18	79.94	89.86	75.39	99.21	**99.91**	81.95	84.30
	Trees	188	1056	1244	92.34	96.69	93.05	96.90	96.78	86.55	**100.00**	99.60
	Soil	186	1056	1242	**98.94**	86.18	98.41	88.44	97.73	96.01	91.29	92.52
	Water	182	143	325	93.09	33.33	85.77	95.72	95.10	90.19	**100.00**	**100.00**
	Residential	196	1072	1268	70.26	**97.09**	63.68	95.25	50.97	79.70	95.80	95.90
	Commercial	191	1053	1244	61.25	71.63	57.90	69.73	65.28	48.34	76.77	**76.91**
	Road	193	1059	1252	75.44	70.93	**78.64**	49.82	54.02	56.89	76.54	76.76
	Highway	191	1036	1227	55.23	72.17	69.95	75.27	77.68	66.20	86.59	**87.44**
	Railway	181	1054	1235	80.17	**85.22**	81.50	70.88	82.45	84.41	74.61	73.95
	Parking-Lot1	192	1041	1233	74.89	63.41	**90.86**	61.17	75.37	48.85	79.89	81.59
	Parking-Lot2	184	285	469	68.53	62.34	81.92	59.92	44.29	68.75	92.12	**92.45**
	Tennis Court	181	247	428	99.02	89.73	**100.00**	88.95	98.34	99.50	**100.00**	**100.00**
	Running Track	187	473	660	97.35	99.36	96.08	94.49	97.78	99.18	**99.58**	99.37
OA (%)					81.25	81.19	81.63	77.42	80.14	76.66	86.09	**86.31**
AA (%)					83.17	79.82	83.06	79.19	81.25	79.34	88.19	**88.52**
Kappa $\times 100$					79.90	79.62	79.63	75.38	78.96	74.78	85.02	**85.15**

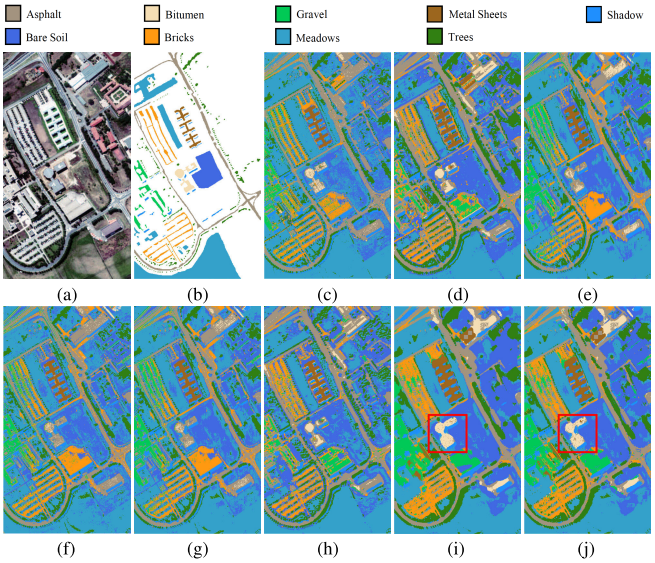


Fig. 8. Classification maps on the Pavia University dataset, acquired from different HIC approaches. (a) False-color Composition. (b) Ground truth. (c) 3-D CNN. (d) GCN. (e) S^2GCN. (f) Nonlocal GCN. (g) MiniGCN. (h) SSAPGCN. (i) ACGT-Net$_{ADD}$. (j) ACGT-Net$_{MUL}$.

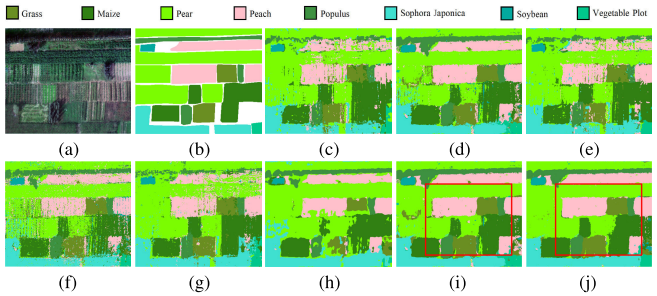


Fig. 9. Classification maps on the Xiong'an Matiwan dataset, acquired from different HIC approaches. (a) False-color Composition. (b) Ground truth. (c) 3-D CNN. (d) GCN. (e) S^2GCN. (f) Nonlocal GCN. (g) MiniGCN. (h) SSAPGCN. (i) ACGT-Net$_{ADD}$. (j) ACGT-Net$_{MUL}$.

accuracies acquired by different approaches. It can be observed in Table III that the proposed ACGT-Net achieves better classification results than other classification methods. The classification maps obtained by different methods are shown in Fig. 8. In Fig. 8, we can observe that the results of our ACGT-Net$_{ADD}$ and ACGT-Net$_{MUL}$ are better than others. The results in Table III and Fig. 8 further demonstrate the effectiveness and competitiveness of our ACGT-Net.

3) Experiment With Xiong'an Matiwan Dataset: As an HSI acquired by a UAV, the Xiong'an dataset has better geometric resolution than other aforementioned datasets. Utilizing the ground truth, we randomly chosen 2% labeled pixels from each class as the training samples. Other hyperparameters follow the settings in the Indian Pines experiment. Table IV provides the classification accuracies obtained from different methods. In Table IV, we can see that the results of our ACGT-Net achieve best classification accuracies. The classification maps obtained by different approaches are illustrated in Fig. 9. As shown in Fig. 9, it can be observed that the classification maps obtained by our ACGT-Net$_{ADD}$ and ACGT-Net$_{MUL}$ are smoother than others. Moreover, the proposed ACGT-Net

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Fig. 10. Classification maps on the Houston University dataset, acquired from different HIC approaches. (a) False-color Composition. (b) Ground truth. (c) 3-D CNN. (d) GCN. (e) S^2GCN. (f) Nonlocal GCN. (g) MiniGCN. (h) SSAPGCN. (i) ACGT-Net$_{\mathrm{ADD}}$. (j) ACGT-Net$_{\mathrm{MUL}}$.

better preserves object details and boundary information than other methods.

4) Experiment With Houston University Dataset: To further verify the effectiveness of our ACGT-Net, we implement different methods on the Houston University dataset. The Houston University dataset is a challenging contest dataset, where the training and test samples are provided by the 2013 IEEE GRSS Data Fusion Contest. The classification accuracies achieved by different approaches are listed in Table V. It can be observed that our ACGT-Net can achieve better results than other methods. Fig. 10 illustrates the classification maps obtained by different methods. In Fig. 10, the results obtained by the ACGT-Net$_{\mathrm{ADD}}$ and ACGT-Net$_{\mathrm{MUL}}$ are smoother than the those of other methods. Obviously, our ACGT-Net can accurately preserve object details and boundary information. Table V and Fig. 10 further verify the effectiveness of our ACGT-Net.

V. Conclusion and Future Lines

In recent years, the development of GCNs opens a new path for HSI processing and stimulates many some new HIC methods. Although these GCN-based methods showed better performance in comparison with conventional methods, their graph structures may not be appropriate for exploiting the advantages of model. Since HSIs are grid data rather than perfect graph data, the structural perturbation problem is unavoidable when using graph model-based networks, which might limit the GNNs' performances on classification. Therefore, refining the graph structures is a practical requirement for applying the theory of GNNs to HIC. In this article, we develop the ACGT-Net for HIC. In the framework of our newly developed ACGT-Net, we first employ the SWMF to aggregate spatial correlations to construct a preliminary graph structure. The training samples are then collected from the structured data to pretrain the GCN as an upstream task, aiming to acquire primary features with local characteristics. Afterward, we design a hybrid domain adaptation to transfer knowledge information from the upstream task to a downstream task. In the downstream task, we adopt the CSS to adaptively refine the graph structure and obtain intermediate features. The CSS-based GSR can mitigate structural perturbations to help the network pay more attention to the most significant features. Finally, the primary and intermediate features are fused as multimodal feature into a softmax layer for classifying nodes and labeling pixels. The significance of our work is that it introduces a meta-heuristic optimization idea to refine graph structure to enhance GCN's performance for HIC. Our future work will investigate and introduce reinforcement learning with meta-heuristic strategies to further enhance classification performance.

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